Gilbert Boadu Ampomah

Los Angeles, CA 90007 | (213)-691-3359
gilbertboadu@outlook.com | ampomah@usc.edu
https://www.linkedin.com/in/gilbert-boadu-ampomah-b27706195/

PROFESSIONAL SUMMARY

Highly motivated researcher specializing in protein chemistry. Experienced in protein semi-synthesis and organic synthesis to investigate the effects of post-translational modifications on disease-relevant proteins *in vitro* and *in cellulo*. Fast learner with strong communication skills; versatile and effective working independently or in collaborative teams.

EDUCATION

University of Southern California (USC), Los Angeles CA, USA Aug 2021–Present
Doctor of Philosophy in Chemistry
Principal Investigator: Prof. Matthew Robert Pratt

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana Sep 2016–Jun 2020
Bachelor of Science in Chemistry, Class Rank: **Top 1%** (5/240)

RESEARCH EXPERIENCE

Pratt Lab, Department of Chemistry, USC Jan 2022–Present
- Recombinantly expressed, purified, and structurally characterized α-synuclein monomers and fibrils from eight animal species using circular dichroism and transmission electron microscopy.
- Established the relationship between α-synuclein sequences from long-lived animal species and aggregation propensity using Thioflavin-T (ThT) aggregation kinetics, sedimentation, and Proteinase-K digestion assays.
- Characterized site-specifically modified HSP27 using recombinant protein expression and solid-phase peptide synthesis to study crosstalk between O-GlcNAcylation and phosphorylation and their effects on protein function and structure.
- Synthesized and validated small molecule probes for specific labeling of O-GlcNAc modified proteins.
- Developed a mammalian cell culture-based assay for distinguishing between enzymatic and chemical labelling of O-GlcNAc modified proteins.

Borquaye Research Group, Department of Chemistry, KNUST Jan 2019–Aug 2021
- Isolated and characterized chemical structures of bioactive plant constituents using chromatographic and spectroscopic tools.
- Tested the biological activities of various plant extracts and isolated active principles using *in vitro* and *in vivo* biochemical assays.
- Executed molecular docking and molecular dynamics simulation of small molecule ligands and protein targets using Chimera, Discovery Studio, PyMol, PyRx, and GROMACS.
- Synthesized and evaluated the biological activity of imidazole derivatives as potential quorum sensing inhibitors.

Laboratory Technician, Central Laboratory, KNUST Nov 2020–Aug 2021
- Analyzed samples using NMR, UV, GC-MS, HPLC, and AAS analytical instruments from across Africa.

RELEVANT SKILLS

- **Molecular Biology:** DNA extraction, Restriction enzyme cloning, Gibson assembly, Site-directed mutagenesis, DNA amplification and purification (PCR), Agarose gel electrophoresis, Mammalian cell culture, Western blotting.
- **Protein Chemistry:** Recombinant protein expression and purification, Protein ligation, High performance liquid chromatography (HPLC), Solid phase peptide synthesis (SPPS), Circular dichroism spectroscopy, Transmission electron microscopy (TEM), Liquid chromatography-mass spectrometry (LC-MS), MALDI-TOF, AKTA-FPLC, Aggregation kinetics assay, Sedimentation assay, Proteinase-K digestion assay, SDS-PAGE, Click chemistry.
- **Organic Synthesis:** Small molecule synthesis, Thin layer chromatography, Column chromatography, Nuclear magnetic resonance (NMR) spectroscopy.

PEER-REVIEWED PUBLICATIONS

8. **Ampomah, G.B.** and Pratt M.R., 2026. Revealing and Exploiting the Biochemistry of O-GlcNAc through Protein Semisynthesis. *Biochemistry*, 65 (7), pp. 801–811.
7. **Ampomah, G.B.**, Hard, E.R. and Pratt, M.R., 2025. α-Synuclein Sequences from Long-Lived Animals Display Generally Diminished Aggregation Compared to Shorter-Lived Animals Including Humans. *ChemBioChem*, *26*(14), p.e202500340.
6. Laryea, M.K., **Ampomah, G.B.**, Ekuadzi, E., Dickson, R.A. and Borquaye, L.S., 2023. Antimalarial compounds from the climbing stems of *Salacia debilis*. *Natural Product Research*, pp.1-10.
5. Aboagye, C.I., **Ampomah, G.B.**, Mensah, J.O., Mensah, C.N., Nartey, D., Gasu, E.N. and Borquaye, L.S., 2023. N-benzylimidazoles as potential antimicrobial and antibiofilm agents–syntheses, bioactivity and molecular docking studies. *Scientific African*, *19*, p.e01529.
4. Mensah, C.N., **Ampomah, G.B.**, Mensah, J.O., Gasu, E.N., Aboagye, C.I., Ekuadzi, E., Boadi, N.O. and Borquaye, L.S., 2022. N-alkylimidazole derivatives as potential inhibitors of quorum sensing in Pseudomonas aeruginosa. *Heliyon*, *8*(12).

3. Mensah, J.O., **Ampomah, G.B.**, Gasu, E.N., Adomako, A.K., Menkah, E.S. and Borquaye, L.S., 2022. Allosteric modulation of the main protease (MPro) of SARS-CoV-2 by casticin—insights from molecular dynamics simulations. *Chemistry Africa*, *5*(5), pp.1305-1320.
2. Kyei, L.K., Gasu, E.N., **Ampomah, G.B.**, Mensah, J.O. and Borquaye, L.S., 2022. An in-silico study of the interactions of alkaloids from *Cryptolepis sanguinolenta* with Plasmodium falciparum dihydrofolate reductase and dihydroorotate dehydrogenase. *Journal of Chemistry*, *2022*(1), p.5314179.
1. Borquaye, L.S., Gasu, E.N., **Ampomah, G.B.**, Kyei, L.K., Amarh, M.A., Mensah, C.N., Nartey, D., Commodore, M., Adomako, A.K., Acheampong, P. and Mensah, J.O., 2020. Alkaloids from *Cryptolepis sanguinolenta* as potential inhibitors of SARS-CoV-2 viral proteins: An in-silico study. *BioMed Research International*, *2020*(1), p.5324560.

CONFERENCES AND PRESENTATIONS

- 5th biennial Chemical Biology and Physiology Conference (December 2025) – Flash talk and poster presentation, Portland, Oregon.
- 52nd annual NOBBChE Conference: Flourishing in scientific, academic, and technical environments" (September 2025) – Poster presentation, Atlanta, Georgia.
- 51st John Stauffer Symposium (April 2025) – Poster presentation, Los Angeles, California.
- Industry days 2025: Industry panel, networking, and poster competition (April 2025) – Poster presentation, Los Angeles, California.
- ACS spring convention (March 2025) – Poster presentation, San Diego.
- International Virtual on Chemistry and Its Applications: Schrodinger Day and International Youth Day, Elsevier Research Building Capacity, by the Department of Chemistry, University of Mauritius (Aug 2020).

TEACHING EXPERIENCE

Burg Teaching Fellow, Department of Chemistry, USC Aug 2025–Dec 2025

- Worked directly with a veteran teaching professor to administer first semester general chemistry lecture course to about 300 students.
- Assisted the professor in the construction and implementation of examinations and other assessments.
- Created and delivered several weeks of lecture materials and the corresponding examinations on covering taught topics.
- Further instructed students in a medium group setting through problem solving sessions.

Teaching Assistant, Department of Molecular and Computational Biology, USC Jan 2025–May 2025

- Supervised cell biology and physiology laboratory sessions.
- Graded tests and assignments for undergraduate students taking cell biology and physiology courses.
- Hosted office hours for partaking undergraduates.
- Set and graded final laboratory examination.

Teaching Assistant, Department of Chemistry, USC Aug 2021–May 2022

- Supervised general chemistry laboratory sessions.
- Graded tests and assignments for undergraduate students taking general chemistry courses.
- Held office hours for undergraduates.

HONORS AND AWARDS

- William Anderson Research Impact Award (2026).
- Anton Burg Pre-doctoral Training Award (2026).
- Anton Burg Pre-doctoral Teaching Fellowship (2025).
- NOBCChE Advancing Science Conference Grant (2025).
- Chemistry Diversity Graduate Fellowship Award (2021).

MEMBERSHIP AND ASSOCIATIONS

- National Organization for the Professional Advancement of Black Chemist and Chemical Engineers (NOBCChE) – Member
- American Chemical Society (ACS) – Member
- Ghana Chemical Society (GCS) – Member

LEADERSHIP POSITIONS

- Chairman of the Academic Committee, Ghana Students' Chemical Society, KNUST (2019–2020).
- Deputy Chairman of the Electoral Committee, Ghana Students' Chemical Society, KNUST (2018–2019).
- Organizing Committee Member, Ghana Students' Chemical Society, KNUST (2017–2018).

PROFESSIONAL REFERENCES

- Matthew Robert Pratt, PhD
  Professor of Chemistry and Departmental Vice Chair, USC
  Phone: 213-740-3014; Email: matthew.pratt@usc.edu
- Jessica Parr, PhD
  Teaching Professor of Chemistry, USC
  Phone: 213-821-6614; Email: parr@usc.edu
- Elias Picazo, PhD
  Assistant Professor of Chemistry, USC
  Phone: 213-740-5976; Email: epicazo@usc.edu